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Comparison of theory and experiments for dispersion in homogeneous porous media

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ABSTRACT

Modeling dispersion in homogeneous porous media with the convection–dispersion equation commonly requires computing effective transport coefficients. In this work, we investigate longitudinal and transverse dispersion coefficients arising from the method of volume averaging, for a variety of periodic, homogeneous porous media over a range of particle Péclet (Pe_p) numbers. Our objective is to validate the upscaled transverse dispersion coefficients and concentration profiles by comparison to experimental data reported in the literature, and to compare the upscaling approach to the more common approach of inverse modeling, which relies on fitting the dispersion coefficients to measured data. This work is unique in that the exact microscale geometry is available; thus, no simplifying assumptions regarding the geometry are required to predict the effective dispersion coefficients directly from theory. Transport of both an inert tracer and non-chemotactic bacteria is investigated for an experimental system that was designed to promote transverse dispersion. We highlight the occurrence of transverse dispersion coefficients that (1) depart from power-law behavior at relatively low Pe_p values and (2) are greater than their longitudinal counterparts for a specific range of Pe_p values. The upscaling theory provides values for the transverse dispersion coefficient that are within the 98% confidence interval of the values obtained from inverse modeling. The mean absolute error between experimental and upscaled concentration profiles was very similar to that between the experiments and inverse modeling. In all cases the mean absolute error did not exceed 12%. Overall, this work suggests that volume averaging can potentially be used as an alternative to inverse modeling for dispersion in homogeneous porous media.

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1. Introduction

Dispersion describes the spreading of solutes within porous media and accounts for the combined effects of diffusion and variations in the convective fluxes within the pores. This phenomenon is generally modeled by a convection–dispersion equation that is applied at the *macroscale*; in other words, it is applied at a scale that is many times larger than the diameter of a single pore. Such macroscale mass balances have been used, often somewhat empirically, for decades. However, recent advances in microscale device manufacturing and in measurement technology have made it possible to rigorously measure and connect the microscale transport phenomena to the effective bulk behavior that is typically measured for such media (*e.g.*, [1–3]). In this work we examine transverse (perpendicular to the direction of the average flow) dispersion in a well-characterized microfluidic device, and compare experimentally-derived values for the transverse dispersion coefficient and concentration profiles with those predicted *a priori* from theory.

Numerous techniques have been developed to estimate dispersion coefficients in porous media on the basis of integrated microscale features (*e.g.*, [1,4–26]). In these techniques, the dispersion coefficients are determined from either (1) sufficient *a posteriori* information about the concentration at locations within the porous medium or (2) appropriately detailed representative *a priori* information about the porous medium microstructure. In the past, methods based on concentration measurements, such as inverse modeling, have been the most practical and popular modeling choices. However, as noted above, recent advances in imaging techniques and computational resources have motivated the use of *predictive* modeling methods that only require representative information about the microstructure of the porous media (*e.g.*, [20,23]). Predictive modeling can be useful in a number of applications; the most important of these, being the potential to actually *design* the structure of the porous medium to promote certain transport behavior. Such an approach could be used, for example, to fine-tune the rate of mass transfer between two regions in a T-sensor microfluidic device, with the idea that the actual rate of mass transfer could be influenced by the particular geometric properties of the pore network within the device. In this work, we provide some evidence that a connection between theory and the experimental device can, in fact, be carried out.

The method of volume averaging is an upscaling technique that only requires representative information about the microstructure of the porous media and involves averaging the governing microscale equations (*i.e.*, the convection–diffusion equation). Briefly, the term

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upscaling refers to a systematic reduction in the number of degrees of freedom involved in the microscale equations resulting in a macroscale model that captures the essential information from the smaller scale [27]. Previous studies (*cf.*, 9–11,14,16,25,28–30) have been devoted to upscaling the convection–diffusion equation in order to provide (*i*) an effective macroscale transport equation, (*ii*) a closure scheme for predicting the total dispersion tensor contained in the macroscale model and (*iii*) a set of length and time scale constraints, which determine the validity of the macroscale model. Here, the terms *microscale* and *macroscale* refer to characteristic lengths on the order of a single pore diameter and hundreds of pore diameters, respectively. In the macroscale model, microscale variations are no longer explicitly apparent, but their influence is represented through the definition of one or more effective medium coefficients and constitutive relationships. These coefficients can be computed from the solution of the associated closure problems, which are responsible for capturing the essential information from the microscale. Fig. 1 illustrates the multi-scale representation of the system investigated in this work. For situations in which there is a small separation of characteristic lengths between the microscale and the macroscale, there is no guarantee that the results from the upscaled model are appropriate. In these situations modified macroscale models have been developed that account for highly localized phenomena and pore-scale non-uniformities (*e.g.*, [31]).

Numerical and experimental investigations have focused on describing the dependence of dispersion coefficients with the particle Péclet number (Pe_p) in two-dimensional ordered media [1,12,15,17–19,22], two-dimensional disordered media [2,14,21,32,33], and three-dimensional media [5,7,8,13,20,23,24,34,35]. The particle Péclet number is defined as (see Chapter 3 in Ref. [30])

$$Pe_p = \frac{\langle \mathbf{v}_\gamma \rangle^\gamma d_p}{\mathscr{D}_A}\left(\frac{\varepsilon_\gamma}{1-\varepsilon_\gamma}\right) \tag{1}$$

where $\langle \mathbf{v}_\gamma \rangle^\gamma$ is the magnitude of the intrinsic averaged velocity vector, d_p is the effective particle diameter, $\mathscr{D}_A$ is the molecular diffusion coefficient and ε_γ is the volume fraction of the fluid phase. Here, the effective particle diameter is defined as $d_p = 6V_p/A_p$ where V_p is the volume of the particles and A_p is the surface area of the particles. With this choice of d_p in Eq. (1), the microscale characteristic length is equivalent to the hydraulic diameter of the porous medium [12].

In many of the above upscaling studies the actual experimental microscale geometry, to which comparisons were made, was idealized using periodic unit cell geometries for the solution of the associated closure problems. To the best of our knowledge, a comparison between upscaling theory and experimental data measured within a periodic porous medium has not been presented in the literature under a volume averaging approach. A comparison of this nature is desirable since the experimental porous system is consistent with one of the simplifying assumptions (*i.e.*, a periodic unit cell) of the upscaling theory. To this end, we use the macroscale convection dispersion equation (arising from the method of volume averaging) to model transport experiments reported by Long and Ford [3] (hereafter, L&F). These experiments consisted of measuring the transverse concentration profiles of an inert tracer and non-chemotactic bacteria within a T-sensor. As shown in Fig. 1, the T-sensor is made of periodic arrays of staggered cylinders of the same diameter. Thus, the closure problems are solved in a unit cell that has the same microstructure as the experimental system, allowing for a detailed analysis of the upscaled model. Additionally, in many cases the comparison between theory and experiments only focused on the predictions of the dispersion coefficients, and not on the evaluation of the predicted macroscale concentration profiles. Thus, we also evaluate the uncertainty regarding the upscaled concentration profiles by comparing them with concentration profiles, and the corresponding 98% confidence intervals, obtained from inverse modeling.

The objectives of this study are (1) to predict the values of longitudinal and transverse dispersion coefficients, arising from the method of volume averaging, for a variety of periodic, homogeneous

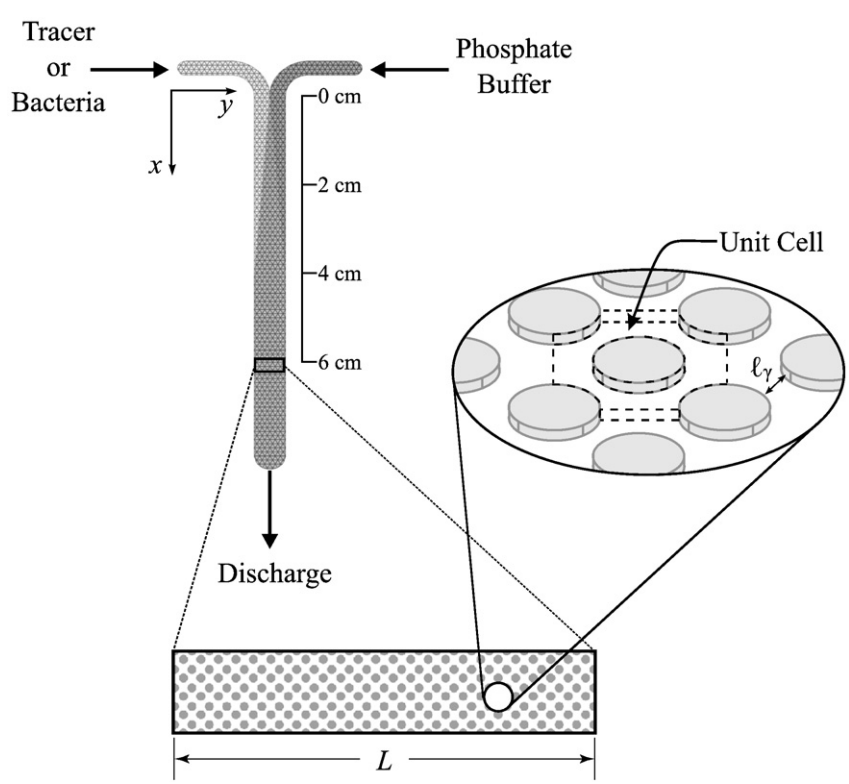

Fig. 1. Length scales and unit cell associated with a T-sensor.

porous media over a range of particle Péclet numbers, (2) to validate the upscaled transverse dispersion coefficients and concentration profiles by comparison to experimental data reported by L&F, and (3) to compare the upscaling approach to the more common approach of inverse modeling, which relies on fitting the dispersion coefficients to measured data. It is acknowledged that the L&F experimental system is highly idealized, however, our comparison between theory and idealized experiments provides detailed quantification of uncertainty in both the upscaling and inverse modeling approaches that is expected to be of importance for general applications of each method.

The remainder of this paper is organized as follows: In Section 2, we first describe the necessary details regarding the experiments performed by L&F. Next, we summarize the volume averaging procedure that results in the macroscale convection–dispersion equation, as well as a closed form expression for the dispersion tensor. Then, the numerical simulations of the closure problem and subsequent macroscale modeling are described. In this work, the closure problem is solved in a 3D unit cell (see Fig. 1), whereas the macroscale model is solved in a vertically averaged, 2D representation of the T-sensor's main channel. We conclude Section 2 with a description of the inverse modeling procedure. In Section 3, we first present the influence of the microstructure on the longitudinal and transverse dispersion coefficients for a range of particle Péclet numbers. There, we highlight the occurrence of transverse dispersion coefficients that (1) depart from power-law behavior at relatively low Pe_p values and (2) are greater than their longitudinal counterparts for a specific range of Pe_p values. Next, we compare the upscaled transverse dispersion coefficients to those obtained by inverse modeling. Lastly, we compare the upscaled and inverse fit concentration profiles to the measured data, as well as present the mean absolute error associated with each method. In Section 4, we discuss our general conclusions and the potential use of volume averaging as an alternative to inverse modeling for dispersion in homogeneous porous media.

2. Methods

2.1. Experimental system

In this work, we focus on the tracer and bacterial control tests that were conducted by L&F. Each test was performed in a microfluidic porous T-sensor (see Fig. 1) with a main channel length of 8.3 cm, a width of 0.6 cm and a depth of 13 μm. The pore structure consisted of staggered cylinders with a radius of 100 μm (separated by pore throats of 46 μm) and a porosity of 0.40. In the tracer test the fluids consisted of 1.2×10^{-4} M fluorecein solution and phosphate buffer, whereas in the bacteria control test the fluids were a bacteria suspension of *E. coli* HCB1 and phosphate buffer (see [3] for details). The molecular diffusion coefficient for fluorecein is $\mathscr{D}_T = 8.2\times10^{-10}\,\mathrm{m^2/s}$ [36] and the random motility coefficient (which is equivalent to molecular diffusion) of the bacteria is $\mathscr{D}_B = 3.0\times10^{-10}\mathrm{m^2/s}$ [37]. In both tests, measurements were taken in the main channel of the T-sensor for mean pore water velocities of 5, 10, and 20 m/day. After steady state was achieved for each mean pore water velocity, images were recorded across three transverse profiles (consisting of 25 pores) at 2, 4, and 6 cm from the head of the main channel. In the tracer tests, three snapshots of fluorecein light were taken at each pore with an epiflourescent microscope (80 ms exposure time) and the fluorescent light was converted to fluorecein concentration. In the bacteria control tests, ten bright field snapshots were taken within a 50×50 μm square region within each pore and the number of bacteria were counted using an ImageJ (NIH) batch script. The total bacteria from the ten snapshots were used to normalize the bacteria concentration within each pore. Thus, in both the tracer and bacteria tests the macroscale experimental transverse concentration profiles consist of concentration estimates within each of the 25 pores along each T-sensor transect.

2.2. Volume averaging

The governing equations at the microscale for convective–diffusive transport of a non-reactive solute are given by

$$\frac{\partial c_{A\gamma}}{\partial t} + \nabla\cdot\left(c_{A\gamma}\mathbf{v}_\gamma\right) = \nabla\cdot\left(\mathscr{D}_A\nabla c_{A\gamma}\right), \text{in the } \gamma\text{−phase} \tag{2a}$$

$$\text{B.C.1}\ \mathbf{n}_{\gamma\kappa}\cdot\left(\mathscr{D}_{A\gamma}\nabla c_{A\gamma}\right) = 0, \text{at } A_{\gamma\kappa} \tag{2b}$$

$$\text{B.C.2}\ c_{A\gamma} = \mathscr{F}_{A\gamma}(\mathbf{r},t), \text{at } A_{\gamma e} \tag{2c}$$

$$\text{I.C.}\ c_{A\gamma} = \mathscr{G}_{A\gamma}(\mathbf{r}), \text{at } t = 0 \tag{2d}$$

where $c_{A\gamma}$ is the concentration of a passive solute, $\mathbf{v}_\gamma$ is the fluid velocity vector and $\mathscr{D}_A$ is the molecular diffusion coefficient. The functions $\mathscr{F}_{A\gamma}(\mathbf{r})$ and $\mathscr{G}_{A\gamma}(\mathbf{r},t)$ (where $\mathbf{r}$ is the position vector) represent the unknown boundary condition at the entrances and exits of the system and the initial condition, respectively. In this study we make the following assumptions: (1) the fluid is incompressible, (2) the flow is fully developed, (3) the velocity field is governed by Stokes equations, and (4) the no-slip boundary condition applies at the fluid–solid interface (denoted by $A_{\gamma\kappa}$). Additionally, Fick's law is assumed to hold for both the tracer and the bacteria. Non-Fickian fluxes, such as those generated by Lévy diffusions, are possible for motile bacteria [38]; however these are not considered in this work.

A drawback of Eq. (2a), especially in studies concerning porous media, is that it is only valid within the fluid phase, thus detailed knowledge regarding the location of the solid phase is required to solve this equation in realistic porous media, which is generally impractical. A more suitable equation, that is valid everywhere within the bulk porous medium, may be written in terms of average quantities and effective medium coefficients as follows (*e.g.*, [39])

$$\frac{\partial\langle c_{A\gamma}\rangle^\gamma}{\partial t} + \langle\mathbf{v}_\gamma\rangle^\gamma\cdot\nabla\langle c_{A\gamma}\rangle^\gamma = \nabla\cdot\left(\mathbf{D}_A^*\cdot\nabla\langle c_{A\gamma}\rangle^\gamma\right). \tag{3}$$

Here $\mathbf{D}_A^*$ is the total (effective) dispersion tensor, whereas $\langle\mathbf{v}_\gamma\rangle^\gamma$ and $\langle c_{A\gamma}\rangle^\gamma$ represent the intrinsic averages of the velocity and the concentration, respectively; these are given explicitly by

$$\langle\mathbf{v}_\gamma\rangle^\gamma = \frac{1}{\mathscr{V}_\gamma}\int_{V_\gamma}\mathbf{v}_\gamma\mathrm{d}V, \quad \langle c_{A\gamma}\rangle^\gamma = \frac{1}{\mathscr{V}_\gamma}\int_{V_\gamma}c_{A\gamma}\mathrm{d}V. \tag{4}$$

In these expressions, V_γ is the region occupied by the fluid phase, and $\mathscr{V}_\gamma$ is its volume. The total dispersion tensor, $\mathbf{D}_A^*$, appearing in Eq. (3) is the sum of the effective diffusivity tensor, $\mathbf{D}_{eff}$, and the hydrodynamic dispersion tensor, $\mathbf{D}_A$,

$$\mathbf{D}_A^* = \mathbf{D}_{eff} + \mathbf{D}_A. \tag{5}$$

Here $\mathbf{D}_{eff}$ and $\mathbf{D}_A$ are defined, respectively, as

$$\mathbf{D}_{eff} = \mathscr{D}_A\left(\mathbf{I} + \frac{1}{\mathscr{V}_\gamma}\int_{A_{\gamma\kappa}}\mathbf{n}_{\gamma\kappa}\mathbf{b}_\gamma\mathrm{d}A\right) \tag{6}$$

$$\mathbf{D}_A = -\langle\tilde{\mathbf{v}}_\gamma\mathbf{b}_\gamma\rangle^\gamma \tag{7}$$

where $\mathbf{I}$ is the identity tensor, and $\tilde{\mathbf{v}}_\gamma = \mathbf{v}_\gamma - \langle\mathbf{v}_\gamma\rangle^\gamma$ are the deviations of the velocity vector. The vector $\mathbf{b}_\gamma$ is the closure variable obtained by solving the following boundary-value problem (BVP) (for details, see [30])

$$\tilde{\mathbf{v}}_\gamma + \mathbf{v}_\gamma\cdot\nabla\mathbf{b}_\gamma = \mathscr{D}_A\nabla^2\mathbf{b}_\gamma \tag{8a}$$

$$\text{B.C.1} - \mathbf{n}_{\gamma\kappa} \cdot \nabla \mathbf{b}_\gamma = \mathbf{n}_{\gamma\kappa}, \text{at } A_{\gamma\kappa} \tag{8b}$$

$$\text{Periodicity } \mathbf{b}_\gamma(\mathbf{r} + \mathbf{l}_i) = \mathbf{b}_\gamma(\mathbf{r}),\ i = 1, 2, 3 \tag{8c}$$

$$\langle \mathbf{b}_\gamma \rangle^\gamma = 0. \tag{8d}$$

Eqs. (8a)–(8d) are formally known as the *closure* problem and are typically solved in a representative periodic region (*i.e.*, a unit cell) that captures the essential physical features of the porous medium microstructure. It should be stressed that the unit cell does not require complete knowledge of the microstructure, but rather assumes that a periodic model of the porous medium is a reasonable representation. It has been shown that a periodic unit cell yields predictions of the effective medium coefficients that are comparable to other theoretical approaches and experimental data in random porous media (*e.g.*, [12,25,30]). In the particular case of the T-sensor investigated here, the medium is in fact periodic, so the assumption of periodicity is not particularly constraining. It is important to point out that in the development of the expressions for $\mathbf{D}_{eff}$ and $\mathbf{D}_A$ (Eqs. (6) and (7)) the length scale constraint, $\ell_\gamma \ll L$, has been assumed in order to simplify the closure problem, where ℓ_γ is a characteristic length of the microscale, and L is a characteristic length of the macroscale.

In this work, the dispersion tensor is assumed to be independent of time and position, which represents the simplest model for the system at hand. Models in which the dispersion tensor evolves in time typically involve non-local closure schemes (*e.g.*, [40–44]) and are not considered here. The total dispersion tensor, $\mathbf{D}_A^*$, is a second rank tensor in which the diagonal components are non-zero and the off-diagonal components are zero when the principal axis is aligned with the average flow direction in anisotropic porous media (*cf.*, [6,39,45,46]). Because the vertical dimension of the microreactor is small compared with the length and width, the averaging process integrates over the entire depth of the reactor. Thus, for the macroscopic-scale transport, only the effective longitudinal and transverse dispersion coefficients (the diagonal components of the tensor) need to be computed. The effective longitudinal dispersion coefficient, $D^*_{A,xx}$, represents dispersion in the direction of the mean flow, whereas the effective transverse dispersion coefficient, $D^*_{A,yy}$ represents dispersion perpendicular to the direction of mean flow in the horizontal plane. Note that this approach does not regard the microscopic domain as two-dimensional. For the computations that follow, the unit cell is taken as a fully represented 3-dimensional volume.

2.3. Numerical simulations

A series of numerical simulations was conducted using the commercial finite-element solver COMSOL Multiphysics 3.4. The linear system solver UMFPACK (included with the COMSOL package) was used for the computation of both the velocity and concentration deviation fields. A standard mesh refining procedure was performed to guarantee that the numerical results were independent of the discretization scheme. Both longitudinal and transverse components of the dispersion tensor were computed in a variety of simple unit cells, as shown in Fig. 2, by solving the closure problem (Eqs. (8a)–(8d)). Here, the simulations were conducted in one half of the full unit cell (illustrated in Fig. 1) since the velocity fields and closure problem solutions are symmetric along the x-axis at the center of the full unit cell. Thus, periodic boundary conditions were imposed at the inlet and outlet of each unit cell half in Fig. 2, whereas symmetry boundary condition were imposed along the sides of the unit cell halves. The symmetry boundary conditions were implemented in COMSOL by setting the diffusive flux to zero. In all unit cell halves, the width was 123 μm, the depth was 13 μm and R_0 was fixed at 100 μm, while maintaining a porosity of 0.40. Each unit cell half consists of two quarter-cylinders of constant radius R_0 and one half-cylinder of variable radius R_1. These six cases correspond to unit cell microstructures in which R_0/R_1 decreases from 5.0 to 1.0. Simulations for which $R_1 > R_0$ were also conducted, however they are not shown in the results since the curves are similar to those obtained for *Case 6*. Note that *Cases 1–5* do not correspond to the geometry of the T-sensor (since $R_1 \neq 100$ μm); these geometries were chosen specifically to investigate the sensitivity of the effective dispersion coefficients on the unit cell microstructure.

The algorithm used for the computation of the closure problem consisted of the following steps:

1. Solve the Stokes and continuity equations for a given pressure gradient in the 3D unit cell halves in Fig. 2.

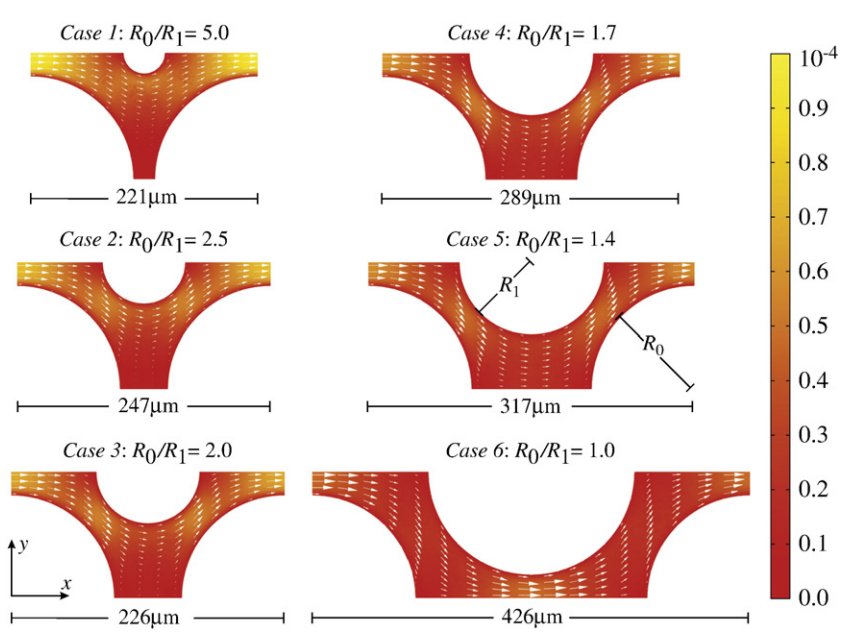

Fig. 2. Top-view of the pore microstructure and flow field ($Pe_p \approx 10$) obtained from numerical solution of the closure problem for several geometries. In all unit cells $R_0 = 100$ μm, the width is 123 μm, and the porosity is 0.40.

2. Using the velocity field from Step 1, solve the x- and y-components of the closure problem given by Eqs. (8a)–(8d).
3. Substitute the results from Steps 1 and 2 into Eqs. (6) and (7) and use them to compute the longitudinal and transverse components of the total dispersion tensor according to Eq. (5).

Case 6 represents the L&F experiments, thus the resulting longitudinal and transverse dispersion coefficients were used to subsequently model a vertically averaged 2D representation of the T-sensor's main channel. In these simulations, the normalized concentration, $C(x,y)$, at $x=0$ cm was taken to be a step change [*i.e.*, $C(0,0\leq y<3\text{ mm})=1$, $C(0,3\leq y\leq 6\text{ mm})=0$] and at the outlet ($x=8.3$ cm) the diffusive flux was set to zero. Fig. 3 illustrates the macroscopic concentration field obtained from these simulations.

2.4. Inverse modeling

The transverse dispersion coefficients for the experiments of L&F were computed from the concentration data by an inverse model. In this approach, the transverse dispersion coefficient was treated as an adjustable parameter and the mean-squared error between the concentration data points and an appropriate analytical solution for the concentration profiles was minimized. The analytical solution describing the transverse concentration profiles is [47]

$$C=\frac{1}{2}\sum_{n=-\infty}^{\infty}\left[\operatorname{erf}\left(\frac{h+2nl-y}{2\sqrt{\mathsf{D}^*_{A,yy}\,x/\langle \mathsf{v}_\gamma\rangle^\gamma}}\right)+\operatorname{erf}\left(\frac{h-2nl+y}{2\sqrt{\mathsf{D}^*_{A,yy}\,x/\langle \mathsf{v}_\gamma\rangle^\gamma}}\right)\right] \tag{9}$$

where C is the normalized concentration, l is the width of the T-sensor channel, and $h=l/2$, while x and y represent the location within the main channel of the T-sensor. It is noted that Eq. (9) was obtained by assuming that transverse dispersion dominates over longitudinal dispersion, while convection is assumed to take place in a semi-infinite domain. Eq. (9) only applies for a small range of Pe_p numbers, which span the Pe_p values for the experiments. In order to obtain a unique dispersion coefficient for each experiment (*i.e.*, for each of the three mean pore water velocities examined), the concentration data measured at each location ($x=2$, 4, and 6 cm) were lumped together for the inverse modeling. For the optimization procedure we used the MATLAB function `lsqcurvefit`, which solves nonlinear curve-fitting problems using least-squares. Sensitivity analysis of the initial guess for the dispersion coefficient indicated that the results are not highly dependent upon the initial input into `lsqcurvefit`. The uncertainty associated with the inverse modeling results was quantified by calculating the 98% confidence interval for each dispersion coefficient. For these calculations standard assumptions regarding the residuals (see [48]) were adopted since they did not exhibit trends suggesting that these assumptions are violated.

3. Results and discussion

3.1. Influence of the microstructure

In this section, $\mathsf{D}^*_{A,xx}$ and $\mathsf{D}^*_{A,yy}$ represent the longitudinal and transverse dispersion coefficients for an arbitrary solute, respectively. These values were obtained by solving the closure problem in the unit cells illustrated in Fig. 2. Fig. 4 shows that for $Pe_p<1$ both $\mathsf{D}^*_{A,xx}$ and $\mathsf{D}^*_{A,yy}$ are minimally affected by the size of the half-cylinder. Moreover, under these conditions $\mathsf{D}^*_{A,xx}\approx\mathsf{D}^*_{A,yy}$ for all cases, which is expected since diffusion controls dispersion almost completely. However, for $Pe_p>10$, Fig. 4 shows that $\mathsf{D}^*_{A,xx}$ decreases as R_1 increases, whereas the opposite is true for $\mathsf{D}^*_{A,yy}$ since the solute deviates from the principal flow direction and travels more laterally (see Fig. 2). The power-law behavior of $\mathsf{D}^*_{A,xx}$ in Fig. 4 is commonly described using $\mathsf{D}^*_{A,xx}/\mathscr{D}_A=aPe_p^b$ where a and b are adjustable parameters (see [30,39], and references therein). The values for a and b for the results presented in Fig. 4 range from 0.002 to 0.016 and from 1.46 to 1.66, respectively, which are consistent with those reported in the literature (*e.g.*, [12,15,19–21,25,30]). Fig. 4 also shows that $\mathsf{D}^*_{A,yy}$ starts to exhibit power-law behavior at Pe_p values in the range of 5–10 for all cases, then as the Pe_p values increase, $\mathsf{D}^*_{A,yy}$ quickly plateaus out of this trend. [19] observed similar results for $\mathsf{D}^*_{A,yy}$ in 2D periodic porous media. The initial power-law behavior is attributed to an increase in lateral convection, however lateral convection is constrained by the geometry of the porous medium and the resulting flow field. The flow field illustrated for each case in Fig. 2 clearly shows that there exists a line of symmetry in every pore of the T-sensor for which the velocity is zero in the transverse direction. Thus, as convection in each pore continues to increase there is less of an increase in transverse dispersion. It is likely that this deviation from power-law behavior for transverse dispersion is entirely the result of the periodic

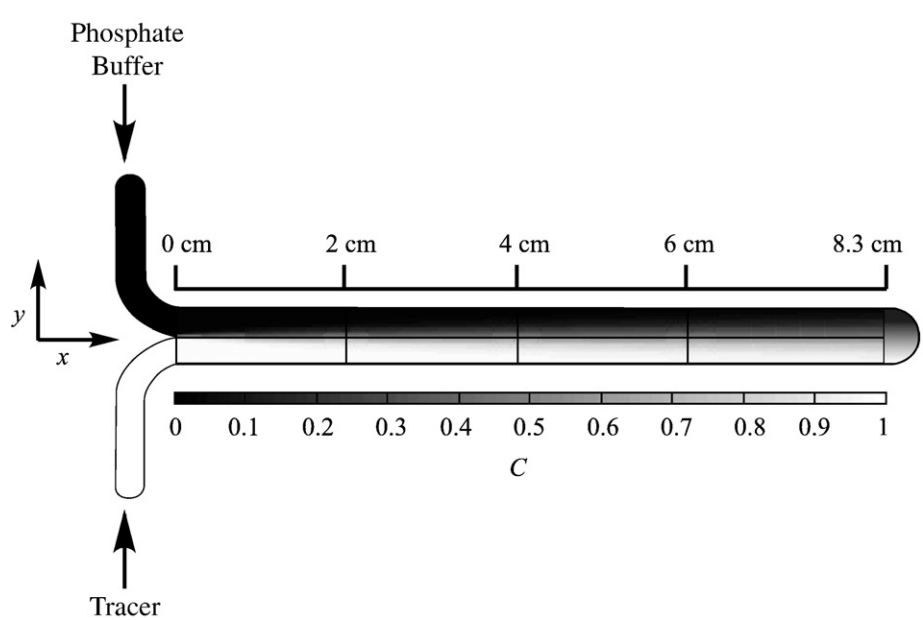

Fig. 3. Example of the 2D macroscale computational domain and the tracer concentration field for the 5 m/day simulation.

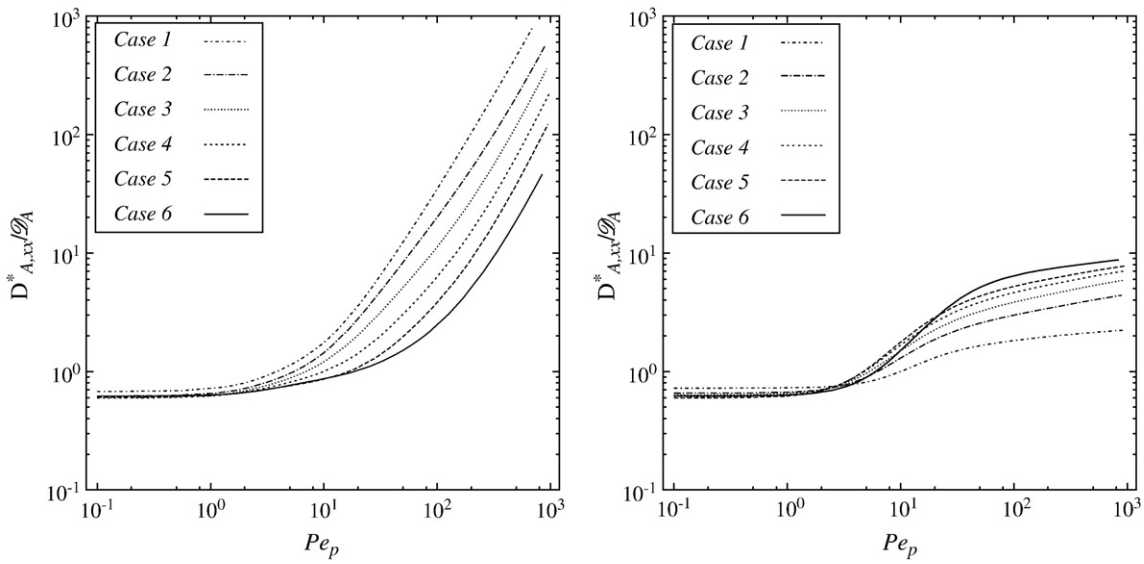

Fig. 4. Normalized longitudinal (left) and transverse (right) dispersion coefficients for *Cases 1–6* obtained from solutions of the closure problem.

structure of the porous medium. Generally, such deviations from the power-law behavior are not observed for randomly packed media (*cf.*, [30]).

In *Cases 3–6*, the unit cell microstructure is such that $D^*_{A,yy}$ is greater than $D^*_{A,xx}$ for different Pe_p number intervals. To illustrate this situation, the results for *Cases 3* and *6* are shown in Fig. 5. These observations are corroborated by [19]. In our simulations, $D^*_{A,yy}$ is greater than $D^*_{A,xx}$ during the transition from the diffusion to the mechanical dispersion regime. In *Case 3*, $D^*_{A,yy}$ is greater than $D^*_{A,xx}$ over a narrow range of Pe_p values ($5<Pe_p<25$). This is due to the existence of a stagnation zone located in-between the two quarter-cylinders where molecular diffusion is still the dominant dispersion mechanism. However, *Case 6* corresponds to a pore geometry in which convection dominates over diffusion almost everywhere in the pores. Consequently, $D^*_{A,yy}$ is greater than $D^*_{A,xx}$ over a wide range of Pe_p values ($5<Pe_p<300$). In this case, there are no large stagnation zones in the pores and convective transport influences transverse dispersion more than in *Cases 3–5*.

3.2. Comparison with experiments

In Fig. 6 we compare the transverse dispersion coefficients arising from inverse modeling with those arising from volume averaging. In all cases, except the tracer experiment at 20 m/day, the upscaled dispersion coefficients are within the 98% confidence interval for the inverse modeling values. Note that the error bars indicate only the uncertainty in the inverse parameter estimation; they do not include additional uncertainty due to the measurement process since these data were not estimated in the original experimental work. The discrepancy observed for the fastest flow rate is almost certainly due to the fact that dispersion is nearing the pre-asymptotic regime for this flow rate. It is interesting to note that the upscaled dispersion coefficients lie on a single curve, which is clearly not the case for the inverse modeling results. For the upscaled results, this was expected since the same physics were assumed for both experimental systems.

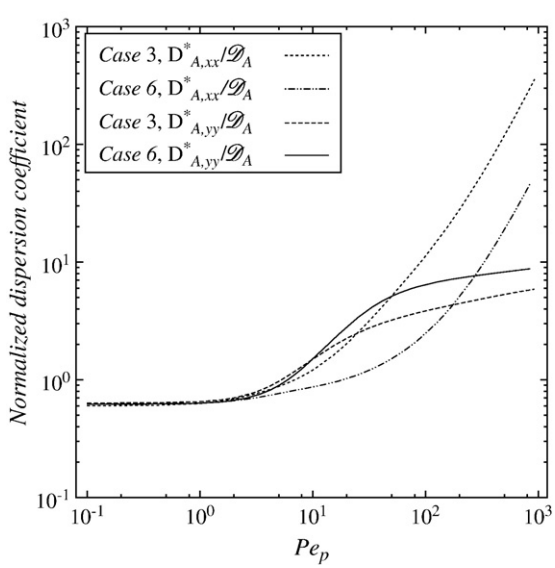

Fig. 5. Normalized dispersion coefficients for *Cases 3* and *6* illustrating that transverse dispersion is greater than longitudinal dispersion for different ranges of Pe_p in each case.

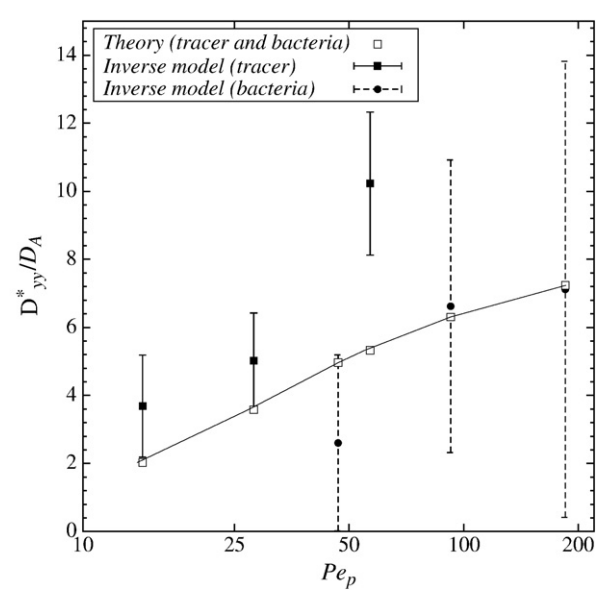

Fig. 6. Comparison of inverse modeling and upscaled (*Case 6*) transverse dispersion coefficients. The error bars on the inverse modeling values represent the 98% confidence interval.

The discrepancies between the tracer and bacteria dispersion coefficients arising from inverse modeling could be caused by any number of factors. L&F argue, among other factors, that the differences between bacteria and tracer particle size could partly explain these differences, but no definitive explanation is provided.

Fig. 7 shows the experimental data, together with the inverse modeling and upscaled concentration profiles for the tracer experiments. At all locations for the 5 and 10 m/day tracer experiments, the upscaled concentration profiles are within the 98% confidence intervals arising from inverse modeling. Essentially, this indicates that the breakthrough curves predicted by the inverse fitting are statistically identical to the forward results developed in this work via the method of volume averaging. In the 20 m/day experiments there are noticeable differences between the inverse and forward approaches. The lack of agreement for our forward model estimates most likely arises because the effective dispersion tensor has not completely relaxed to its asymptotic value. As mentioned above, for high flow rates it may be necessary to consider a *pre-asymptotic* (*i.e.*, time varying) dispersion tensor; although this is beyond the scope of the present effort, it is an interesting problem that deserves further study. Table 1 shows that the mean absolute errors (MAE) between the upscaled and experimental concentration profiles for the tracer are on the order of 10%, which are comparable to the MAE between the inverse modeling and the experimental data. Moreover, in all cases the MAE for inverse modeling and upscaling are very similar with the largest difference of 0.4% occurring in the 5 m/day experiment at 6 cm.

Table 1
Percent of mean absolute error for the tracer experiments.

	5 m/day		10 m/day		20 m/day	
	IM	VA	IM	VA	IM	VA
2 cm	5.3	5.3	6.4	6.2	8.1	7.9
4 cm	7.3	7.1	6.1	6.0	6.1	6.1
6 cm	12.2	11.8	6.0	5.9	8.8	8.7

IM = inverse modeling, VA = volume averaging.

Fig. 8 shows the comparison between the experimental, inverse modeling and upscaled concentration profiles for the bacteria experiments. The upscaled concentration profiles are within the 98% confidence intervals at all locations for each flow rate. Table 2 shows that in most cases the MAE associated with inverse modeling and upscaling for bacteria are identical (to one decimal place). In the cases for which the MAE is not the same, the difference is 0.1%. The comparisons in Figs. 6 and 7 clearly show that the concentration

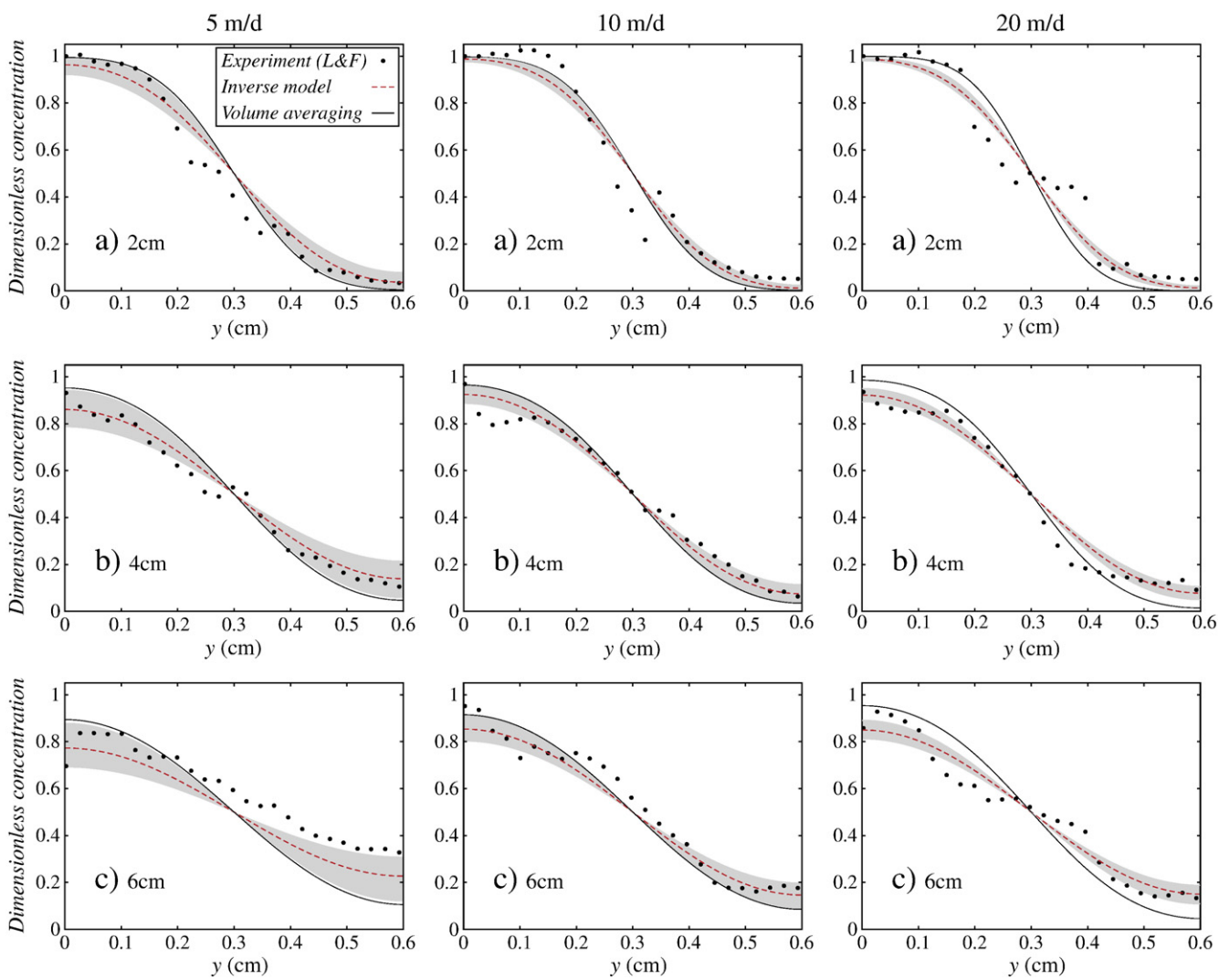

Fig. 7. Comparison of the experimental, inverse modeling and upscaled concentration profiles for the tracer experiments. The transverse concentration profiles span the entire width of the T-sensor main channel at a) $x = 2$ cm, b) $x = 4$ cm, and c) $x = 6$ cm (see Fig. 3). The grey regions represent the 98% confidence intervals associated with the inverse modeling dispersion coefficients.

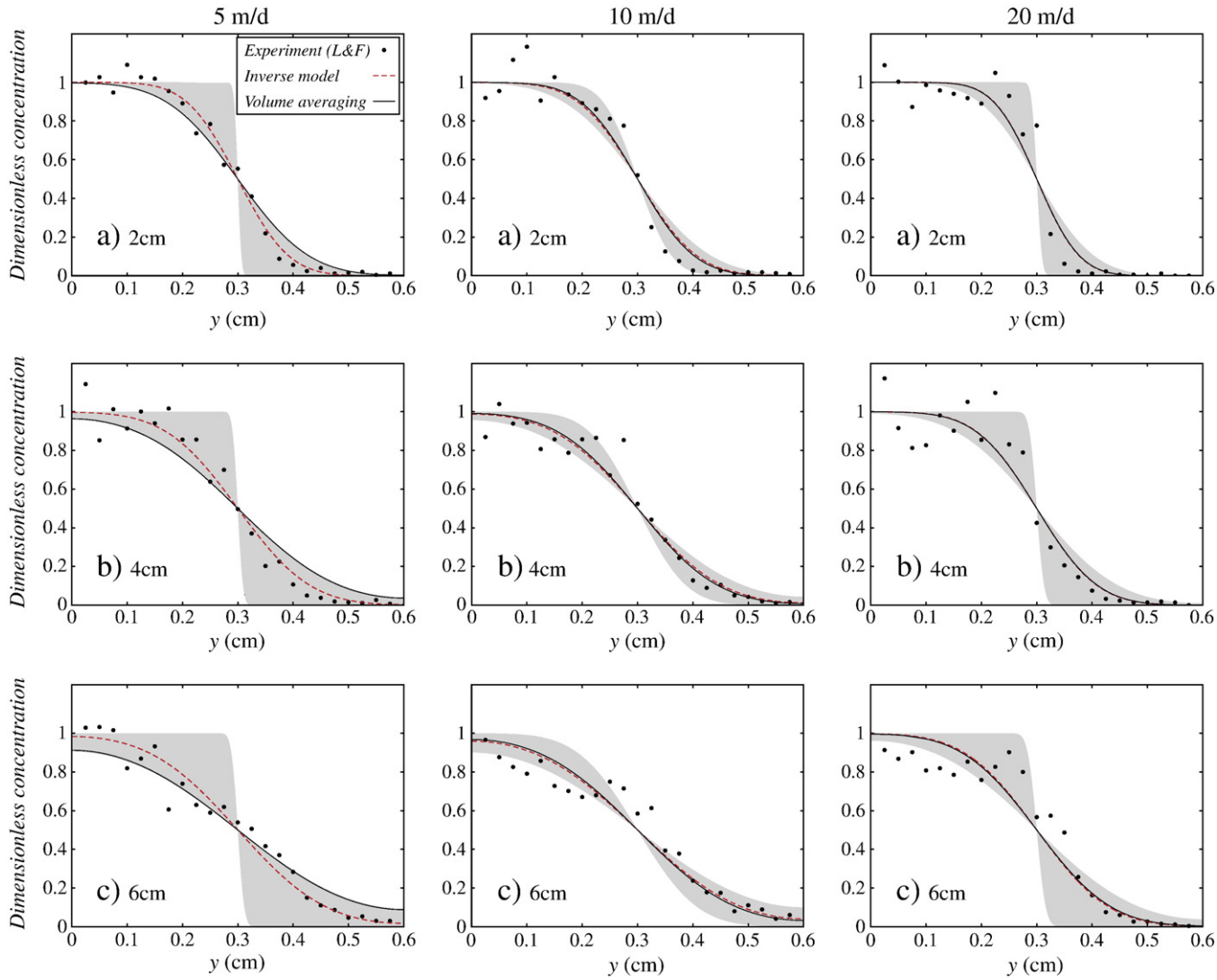

Fig. 8. Comparison of the experimental, inverse modeling and upscaled concentration profiles for the bacteria experiments. The transverse concentration profiles span the entire width of the T-sensor main channel at a) $x = 2$ cm, b) $x = 4$ cm, and c) $x = 6$ cm (see Fig. 3). The grey regions represent the 98% confidence intervals associated with the inverse modeling dispersion coefficients.

profiles arising from volume averaging are reliable predictions of the measured concentration data.

4. Summary and conclusions

We applied the method of volume averaging to dispersion in a periodic, homogeneous porous media. The work presented here deals with the simplest conceptual model for dispersion, in that the dispersion coefficient is assumed to be independent of time and position. We investigated the role of the microstructure on the effective medium coefficients and found that, under certain conditions, the transverse dispersion coefficient can be greater than the longitudinal dispersion coefficient. Moreover, we evaluated the predictive capabilities of volume averaging by comparing the dispersion coefficients and concentration profiles arising from upscaling with the experimental data and inverse modeling. Overall, the upscaled dispersion coefficients and concentration profiles were not appreciably different than those obtained from the inverse modeling procedure. These results are encouraging and of practical value since the two modeling approaches considered here require different information about the system. On the one hand, the inverse modeling approach requires experimental data as input and treats the effective medium coefficients as adjustable parameters that are optimized to fit the measured data. On the other hand, the method of volume averaging is a predictive approach that only requires representative (and not necessarily detailed) information about the porous medium without the use of adjustable coefficients.

Further validation studies and applications of the method of volume averaging to more complex modeling problems are required in order to draw general conclusions regarding the applicability of volume averaging as a predictive tool. Work is currently underway that addresses time and position dependent dispersion coefficients in simple and more complex porous systems.

Table 2
Percent of mean absolute error for the bacteria experiments.

	5 m/day		10 m/day		20 m/day	
	IM	VA	IM	VA	IM	VA
2 cm	4.5	4.5	5.8	5.7	6.2	6.2
4 cm	8.2	8.2	5.1	5.1	7.4	7.4
6 cm	6.7	6.6	7.6	7.5	8.2	8.2

IM = inverse modeling, VA = volume averaging.

Notation

$A_{\gamma\sigma}$	area of the fluid–solid interface, m^2
A_p	surface area of the solid phase, m^2
$\mathbf{b}_\gamma$	closure variable that maps $\nabla\langle c_{A\gamma}\rangle$ onto $\tilde{c}_{A\gamma}$, m
$c_{A\gamma}$	microscale concentration, mol/m^3
$\langle c_{A\gamma}\rangle^\gamma$	intrinsic average of the concentration, mol/m^3
C	normalized macroscale concentration
d_p	effective particle diameter, m
$\mathscr{D}_A$	molecular diffusion coefficient, m^2/s
$\mathscr{D}_B$	random motility coefficient of the bacteria, m^2/s
$\mathscr{D}_T$	molecular diffusion coefficient of the tracer, m^2/s
$D^*_{A,xx}$	longitudinal dispersion coefficient, m^2/s
$D^*_{A,yy}$	transverse dispersion coefficient, m^2/s
$\mathbf{D}_A$	hydrodynamic dispersion tensor, m^2/s
$\mathbf{D}^*_A$	total dispersion tensor, m^2/s
$\mathbf{D}_{eff}$	effective diffusion tensor, m^2/s
h	half the width of the main channel of the T-sensor, m
$\mathbf{I}$	identity tensor
ℓ_γ	characteristic length associated to the γ-phase, m
l	width of the main channel of the T-sensor, m
$\mathbf{l}_i$	unit cell lattice vectors, m
L	characteristic length associated to the macroscale, m
$\mathbf{n}_{\gamma\sigma}$	unit normal vector directed from the γ-phase toward the σ-phase
Pe_p	particle Péclet number
$\mathbf{r}$	position vector, m
R_0	radius of the two large cylinders in the unit cell, μm
R_1	radius of the small cylinder in the unit cell, μm
t	time, s
$\mathbf{v}_\gamma$	fluid velocity vector, m/s
$\langle\mathbf{v}_\gamma\rangle^\gamma$	intrinsic average of the fluid velocity vector, m/s
$\tilde{\mathbf{v}}_\gamma$	spatial deviations of the fluid velocity, m/s
$\langle\mathbf{v}_\gamma\rangle^\gamma$	magnitude of the intrinsic averaged velocity vector, m/s
V	averaging region
$\mathscr{V}$	volume of the averaging region, m^3
$\mathscr{V}_\gamma$	volume of the γ-phase contained within the averaging region, m^3
V_p	volume of the solid phase, m^3

Greek symbols

ε_γ	volume fraction (porosity) of the fluid phase

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